

Separation of the Divergent and Finite Parts of the LO Trijet Cross-Section

Direct subtraction of the slow-gluon pole — no phase-space slicing, no cutoffs

Worked out directly on `Trijet_L0.pdf` ("Trijet divergences", §1.1–§1.3). All equation numbers of the form (1.n) refer to that document; its notation is used throughout without change.

Every algebraic step below is verified symbolically; the scripts are in `trijet/checks/`.

0. Result

Your trijet cross-section is a **fully differential** three-parton distribution: all three momenta k_1, k_2, k_3 are fixed, so there is *no* phase-space integration to slice. The whole singular structure sits in the explicit $1/\xi$ of the prefactors of (1.1), (1.9) and (1.17), where $\xi = k_3^+/q^+$ is the gluon light-cone momentum fraction. Everything else is finite — in particular the transverse integrals are already ultraviolet-convergent, because the four-term $\mathcal{W}$ combination supplies its own subtraction (§2.2).

So the separation is the standard one: **add and subtract the $\xi \rightarrow 0$ limit of the integrand**.

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} = \frac{1}{\xi} \mathcal{G}(\xi) = \underbrace{\frac{\mathcal{G}(0)}{\xi}}_{\text{divergent}} + \underbrace{\frac{\mathcal{G}(\xi) - \mathcal{G}(0)}{\xi}}_{\text{finite}}, \quad \mathcal{G}(\xi) - \mathcal{G}(0) = \mathcal{O}(\xi).$$

Equivalently, as a distribution on $\xi \in (0, \xi_f]$ with $\xi_0 = \Lambda^+/q^+$ the rapidity regulator and ξ_f the rapidity factorization scale,

$$\frac{1}{\xi} \Big|_{\xi > \xi_0} = \left[\frac{1}{\xi} \right]_+^{(\xi_f)} + \delta(\xi) \ln \frac{\xi_f}{\xi_0}, \quad \int_0^{\xi_f} d\xi \left[\frac{1}{\xi} \right]_+^{(\xi_f)} f(\xi) \equiv \int_0^{\xi_f} \frac{d\xi}{\xi} [f(\xi) - f(0)].$$

The content of the calculation is $\mathcal{G}(0)$, computed exactly in §3. The answer is

$$\mathcal{G}(0) = \mathcal{C} \, 16 \, \zeta^2 \bar{\zeta}^2 [\zeta^2 + \bar{\zeta}^2] \int_{\Xi} (R_0 \cdot \bar{R}_0) \frac{K_1(Q\mathcal{D}_0) K_1(Q\bar{\mathcal{D}}_0)}{\mathcal{D}_0 \bar{\mathcal{D}}_0} \left\{ \mathcal{K}_{\bar{q}\bar{q}} \mathbb{W}_{\bar{q}}^{(0)} + \mathcal{K}_{qq} \mathbb{W}_q^{(0)} - 2 \Re[\mathcal{K}_{q\bar{q}} \mathbb{W}_{q\bar{q}}^{(0)}] \right\}$$

with $\bar{\zeta} \equiv 1 - \zeta$, the three eikonal kernels

$$\mathcal{K}_{\bar{q}\bar{q}} = \frac{(z - y) \cdot (\bar{z} - \bar{y})}{(z - y)^2 (\bar{z} - \bar{y})^2}, \quad \mathcal{K}_{qq} = \frac{(z - x) \cdot (\bar{z} - \bar{x})}{(z - x)^2 (\bar{z} - \bar{x})^2}, \quad \mathcal{K}_{q\bar{q}} = \frac{(z - x) \cdot (\bar{z} - \bar{y})}{(z - x)^2 (\bar{z} - \bar{y})^2},$$

and the three Wilson-line structures, all of which collapse to pure quadrupole/dipole products — every $S_{q\bar{q}g}$ and every +1 in (1.5)–(1.8), (1.13)–(1.16), (1.21)–(1.24) cancels:

$$\begin{aligned} \mathbb{W}_{\bar{q}}^{(0)} &= Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - Q(x, y, \bar{z}, \bar{x}) S(\bar{y}, \bar{z}) - Q(x, z, \bar{y}, \bar{x}) S(z, y) + Q(x, y, \bar{y}, \bar{x}), \\ \mathbb{W}_q^{(0)} &= Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z}) - Q(y, x, \bar{z}, \bar{y}) S(\bar{x}, \bar{z}) - Q(y, z, \bar{x}, \bar{y}) S(z, x) + Q(y, x, \bar{x}, \bar{y}), \\ \mathbb{W}_{q\bar{q}}^{(0)} &= Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - Q(x, y, \bar{y}, \bar{z}) S(\bar{z}, \bar{x}) - Q(x, z, \bar{y}, \bar{x}) S(z, y) + S(x, y) S(\bar{y}, \bar{x}). \end{aligned}$$

This is **exactly the real-emission B-JIMWLK kernel at large N_c acting on the LO $\gamma_T^* \rightarrow q\bar{q}$ dijet**. The check is in §4.2: at coincident transverse points the three kernels combine into the BK dipole kernel, $\mathcal{K}_{\bar{q}\bar{q}} + \mathcal{K}_{qq} - 2\mathcal{K}_{q\bar{q}} = \frac{(\mathbf{x} - \mathbf{y})^2}{(\mathbf{z} - \mathbf{x})^2 (\mathbf{z} - \mathbf{y})^2}$.

1. Notation and the three inputs

1.1 Momentum fractions

From $\delta(q^+ - k_1^+ - k_2^+ - k_3^+)$ and from the way $(1 - \zeta)$ appears in (1.2)–(1.4):

$$\zeta = \frac{k_1^+}{q^+} \text{ (quark)}, \quad \eta = \frac{k_2^+}{q^+} \text{ (antiquark)}, \quad \xi = \frac{k_3^+}{q^+} \text{ (gluon)}, \quad \zeta + \eta + \xi = 1,$$

so that $\bar{\zeta} \equiv 1 - \zeta = \eta + \xi$ is the fraction carried by the $\bar{q}g$ system before the emission, and $1 - \eta = \zeta + \xi$ that of the qg system. This reading is forced by (1.2): the factor $[\zeta^2 + (1 - \zeta)^2]$ is the $\gamma_T^* \rightarrow q\bar{q}$ splitting weight with the quark taking ζ , and $[\eta^2 + (1 - \zeta)^2]$ is the $\bar{q} \rightarrow \bar{q}g$ weight with the antiquark going from $1 - \zeta$ to η .

1.2 Compact form of your three contributions

Write

$$\mathcal{C} \equiv \frac{e^2 g^2 Q^2 C_F N_c}{8 (2\pi)^{12} (q^+)^2} \left(\sum_f e_f^2 \right) \delta(q^+ - k_1^+ - k_2^+ - k_3^+), \quad \int_{\Xi} \equiv \int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-ik_1 \cdot (x-\bar{x})} e^{-ik_2 \cdot (y-\bar{y})} e^{-ik_3 \cdot (z-\bar{z})}.$$

Then (1.1), (1.9), (1.17) read

$$\begin{aligned} \frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} &= \frac{\mathcal{C}}{\xi} \left[\mathcal{I}_{\bar{q}} + \mathcal{I}_q - 2 \Re \mathcal{I}_{q\bar{q}} \right] \equiv \frac{\mathcal{G}(\xi)}{\xi}, \\ \mathcal{I}_{\bar{q}} &= \zeta^2 \int_{\Xi} \frac{R^i Y^m \bar{R}^j \bar{Y}^n}{Y^2 \bar{Y}^2} \left[\Phi_{\lambda_1 \lambda}^{lirm}(x, y, z, \zeta, \xi) \Phi_{\lambda_1 \lambda}^{ljrn*}(\bar{x}, \bar{y}, \bar{z}, \zeta, \xi) \right] \frac{K_1(Q\mathcal{D}) K_1(Q\bar{\mathcal{D}})}{\mathcal{D}\bar{\mathcal{D}}} \mathbb{W}_{\bar{q}}, \\ \mathcal{I}_q &= \eta^2 \int_{\Xi} \frac{\tilde{R}^i X^m \tilde{R}^j \bar{X}^n}{X^2 \bar{X}^2} \left[\tilde{\Phi} \tilde{\Phi}^* \right] \frac{K_1(Q\mathcal{D}) K_1(Q\bar{\mathcal{D}})}{\mathcal{D}\bar{\mathcal{D}}} \mathbb{W}_q, \quad \mathcal{I}_{q\bar{q}} = \eta \zeta \int_{\Xi} \frac{\tilde{R}^i X^m \bar{R}^j \bar{Y}^n}{X^2 \bar{Y}^2} \left[\tilde{\Phi} \Phi^* \right] \frac{K_1 K_1}{\mathcal{D}\bar{\mathcal{D}}} \mathbb{W}_{q\bar{q}}, \end{aligned}$$

with the four-term Wilson-line combinations

$$\begin{aligned} \mathbb{W}_{\bar{q}} &= \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) - \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') + \mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}'), \\ \mathbb{W}_q &= \mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - \mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) - \mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') + \mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}'), \\ \mathbb{W}_{q\bar{q}} &= \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) - \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') + \mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}'). \end{aligned}$$

1.3 The two properties of y', x' that are used

$y' (x')$ is the transverse position of the $\bar{q}g (qg)$ parent, i.e. the light-cone centroid

$$\mathbf{y}' = \frac{\eta \mathbf{y} + \xi \mathbf{z}}{\eta + \xi}, \quad \mathbf{x}' = \frac{\zeta \mathbf{x} + \xi \mathbf{z}}{\zeta + \xi}.$$

Only two consequences are needed below, and both are independent of the precise weights:

- (P1) $\mathbf{y}' \rightarrow \mathbf{y}$ and $\mathbf{x}' \rightarrow \mathbf{x}$ as $\xi \rightarrow 0$ — used for the soft limit (§3.2).
(P2) $\mathbf{y}' \rightarrow \mathbf{y}$ as $\mathbf{z} \rightarrow \mathbf{y}$, and $\mathbf{x}' \rightarrow \mathbf{x}$ as $\mathbf{z} \rightarrow \mathbf{x}$, with $\mathbf{y}' - \mathbf{y} = \frac{\xi}{\eta + \xi}(\mathbf{z} - \mathbf{y})$ — used for the UV finiteness (§2.2).

1.4 The soft limits of $\mathcal{D}, R, Y, X, \tilde{R}$ — stated as assumptions

These objects are not defined in the excerpt, so I use only that they are continuous at $\xi = 0$ and reduce there to the corresponding leading-order and eikonal objects:

- (P3) $\mathcal{D}(x, y, z, \zeta, \xi)|_{\xi=0} = \mathcal{D}(y, x, z, \eta, \xi)|_{\xi=0} \equiv \mathcal{D}_0(x, y, \zeta)$, the LO $q\bar{q}$ effective size.
(P4) $R|_{\xi=0} = \tilde{R}|_{\xi=0} \equiv R_0(x, y, \zeta)$, the LO transverse vector.
(P5) $\frac{Y^m}{Y^2}|_{\xi=0} = \frac{(\mathbf{z} - \mathbf{y})^m}{(\mathbf{z} - \mathbf{y})^2}, \quad \frac{X^m}{X^2}|_{\xi=0} = \frac{(\mathbf{z} - \mathbf{x})^m}{(\mathbf{z} - \mathbf{x})^2}.$

These are checkable, not free. (P3)–(P4) are exactly what is required for the three contributions to share a common prefactor, and (P5) is what turns the sum into the BK kernel — see §4.2. If your $\mathcal{D}, R, Y, X$ satisfy them (they will, if $\mathcal{D}^2 = \zeta \eta \mathbf{r}_{xy}^2 + \zeta \xi \mathbf{r}_{xz}^2 + \eta \xi \mathbf{r}_{yz}^2$ and

$\mathbf{Y} \propto \mathbf{z} - \mathbf{y}$ up to $\mathcal{O}(\xi)$, which is the standard LCPT form), everything below is exact. If not, only §3.3 changes; §3.1 and §3.2 are unconditional.

2. Where the singularities are — complete survey

2.1 The $1/\xi$ pole: the only genuine divergence

All three prefactors in (1.1), (1.9), (1.17) carry $1/\xi$. Since $\mathcal{G}(\xi)$ has a finite, non-vanishing limit as $\xi \rightarrow 0$ (computed in §3), the cross-section behaves as

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} \xrightarrow{\xi \rightarrow 0} \frac{\mathcal{G}(0)}{\xi},$$

a simple pole. Its integral over ξ diverges logarithmically at the lower end, which is the familiar **rapidity (small- x , "slow gluon") divergence**. It is regulated by a cutoff on the gluon's plus momentum, $k_3^+ > \Lambda^+$, i.e. $\xi > \xi_0 = \Lambda^+/q^+$.

2.2 The transverse ultraviolet is already subtracted — no $1/\epsilon$ anywhere

The kernels Y^m/Y^2 and $\bar{Y}^n/\bar{Y}^2$ behave as $1/|\mathbf{z} - \mathbf{y}|$ as $\mathbf{z} \rightarrow \mathbf{y}$, which would make $\int d^2 z d^2 \bar{z}$ logarithmically divergent. It is not, because the four-term $\mathcal{W}$ combination vanishes there. Using (P2), at $\mathbf{z} = \mathbf{y}$ one has $\mathbf{y}' = \mathbf{y}$ identically, so the first and second terms of $\mathbb{W}_{\bar{q}}$ coincide, as do the third and fourth:

$$\mathbb{W}_{\bar{q}} \Big|_{\mathbf{z} \rightarrow \mathbf{y}} = 0, \quad \mathbb{W}_{\bar{q}} \Big|_{\bar{\mathbf{z}} \rightarrow \bar{\mathbf{y}}} = 0, \quad \mathbb{W}_q \Big|_{\mathbf{z} \rightarrow \mathbf{x}} = 0, \quad \mathbb{W}_{q\bar{q}} \Big|_{\mathbf{z} \rightarrow \mathbf{x}} = 0, \quad \text{etc.}$$

Quantitatively, $\mathbf{y}' - \mathbf{y} = \frac{\xi}{\eta + \xi}(\mathbf{z} - \mathbf{y})$, so

$$\mathbb{W}_{\bar{q}} \simeq \frac{\xi}{\eta + \xi} \frac{\xi}{\eta + \xi} (\mathbf{z} - \mathbf{y})^a (\bar{\mathbf{z}} - \bar{\mathbf{y}})^b \partial_{y^a} \partial_{\bar{y}^b} \mathcal{W} + \dots = \mathcal{O}(|\mathbf{z} - \mathbf{y}| |\bar{\mathbf{z}} - \bar{\mathbf{y}}|),$$

which exactly compensates the $1/(|\mathbf{z} - \mathbf{y}| |\bar{\mathbf{z}} - \bar{\mathbf{y}}|)$ of the kernels. The integrand is therefore bounded and $\int d^2 z d^2 \bar{z}$ converges. **No dimensional regularization and no UV counterterm are required:** your $\mathcal{W}$ organization already is the subtraction.

The "regular $\times$ instantaneous" structure (1.3) carries an extra $Y^2/(R \cdot Y) \sim |\mathbf{Y}|$ and is *less* singular; "instantaneous $\times$ instantaneous" (1.4) carries $Y^2 \bar{Y}^2/[(R \cdot Y)(\bar{R} \cdot \bar{Y})]$ and is finite by itself. So the statement above, established for the most singular structure, covers all of them.

2.3 The collinear region is a boundary, not a pole

At fixed, generic, non-degenerate k_1, k_2, k_3 there is no collinear singularity: the configuration $k_3/\xi \rightarrow k_2/\eta$ is a point on the boundary of the three-parton phase space, not a singularity of the integrand. A collinear logarithm appears only if you subsequently *integrate* over the relative momentum of two of the partons — i.e. only when you turn the trijet into a dijet. See §7.

2.4 Large $|\mathbf{z}|$

Colour transparency: as the gluon is taken far from the $q\bar{q}$ pair, the quadrupoles factorize and $\mathbb{W} \rightarrow 0$; in addition the kernels fall as $1/z^2$ and the phase $e^{-ik_3 \cdot (z - \bar{z})}$ oscillates at fixed non-zero k_3 . No infrared divergence.

2.5 Endpoints in ζ, η

(1.3), (1.4) carry $1/(1 - \zeta)$ and (1.19) carries $1/(\zeta + \xi)$. Since $1 - \zeta = \eta + \xi$ and $\zeta + \xi = 1 - \eta$, these blow up only when the antiquark *and* the gluon (resp. the quark and the gluon) both become soft. At fixed jet momenta they are regular. Note also that they always appear multiplied by at least one power of ξ — $\frac{8\xi\eta}{1-\zeta}$ in (1.3), $\left[\frac{\xi\zeta}{1-\zeta}\right]^2$ in (1.4), $\frac{8\xi\zeta}{\zeta+\xi}$ in (1.19) — so they do not survive the soft limit either.

Limit	Potential singularity	Status
$\xi \rightarrow 0$ (slow gluon)	$1/\xi$ pole	The divergence. Subtracted in §5, absorbed into B-JIMWLK/BK.
$z \rightarrow y, z \rightarrow x$	$\log UV$ of $\int d^2z$	Cancels identically inside $\mathbb{W}$ (§2.2).
$ z \rightarrow \infty$	infrared	Regulated by colour transparency.
$k_3/\xi \rightarrow k_2/\eta$	collinear	Boundary of phase space; not a pole at fixed generic momenta.
$\zeta \rightarrow 1, \eta \rightarrow 0$	endpoint	Always accompanied by explicit powers of ξ ; regular.

3. The soft limit $\mathcal{G}(0)$, computed exactly

3.1 The spinor structures

Substitute $\eta = 1 - \zeta - \xi$ in (1.2)–(1.4), (1.10)–(1.12), (1.18)–(1.20) and let $\xi \rightarrow 0$. Every structure carrying an explicit power of ξ dies, which removes *all* the $\epsilon^{ij}\epsilon^{mn}$ (helicity-flip) terms and both instantaneous structures:

Structure	Eq.	Explicit ξ factor	$\xi \rightarrow 0$
$\text{Reg} \times \text{Reg}, \delta^{ij}\delta^{mn}$	(1.2),(1.10),(1.18)	none	survives
$\text{Reg} \times \text{Reg}, \epsilon^{ij}\epsilon^{mn}$	(1.2),(1.10),(1.18)	ξ	0
$\text{Reg} \times \text{Inst} + \text{h.c.}$	(1.3),(1.11),(1.19)	ξ	0
$\text{Inst} \times \text{Inst}$	(1.4),(1.12)	ξ^2	0
$\text{Inst} \times \text{Inst}$ (interference)	(1.20)	—	0 identically

So only the $\delta^{ij}\delta^{mn}$ piece of $\text{Regular} \times \text{Regular}$ survives. Including the prefactors of (1.1), (1.9), (1.17), define

$$\begin{aligned}
\mathcal{A}_{\bar{q}}(\zeta, \xi) &\equiv \zeta^2 [\Phi \Phi^*]_{\delta\delta} = 8\zeta^2 [\zeta^2 + \bar{\zeta}^2] (\eta^2 + \bar{\zeta}^2) = 8\zeta^2 [\zeta^2 + \bar{\zeta}^2] (2\bar{\zeta}^2 - 2\bar{\zeta}\xi + \xi^2), \\
\mathcal{A}_q(\zeta, \xi) &\equiv \eta^2 [\tilde{\Phi} \tilde{\Phi}^*]_{\delta\delta} = 8(\bar{\zeta} - \xi)^2 [\zeta^2 + (\zeta + \xi)^2] [(\bar{\zeta} - \xi)^2 + (\zeta + \xi)^2], \\
\mathcal{A}_{q\bar{q}}(\zeta, \xi) &\equiv \eta\zeta [\tilde{\Phi} \Phi^*]_{\delta\delta} = 8\zeta(\bar{\zeta} - \xi) [1 - \xi - 2\zeta(\bar{\zeta} - \xi)] [\xi(1 - \xi) + 2\zeta(\bar{\zeta} - \xi)].
\end{aligned}$$

Evaluating at $\xi = 0$, and using $1 - 2\zeta\bar{\zeta} = \zeta^2 + \bar{\zeta}^2$:

$$\mathcal{A}_{\bar{q}}(\zeta, 0) = \mathcal{A}_q(\zeta, 0) = \mathcal{A}_{q\bar{q}}(\zeta, 0) = 16\zeta^2\bar{\zeta}^2[\zeta^2 + \bar{\zeta}^2] \equiv \mathcal{A}_0(\zeta)$$

All three channels collapse to the *same* weight. That is not an accident: it is the statement that a slow gluon does not resolve which parton emitted it, and it is what allows the three transverse kernels to combine (§4.2). [Verified symbolically, `checkA.py`.]

Note that $\mathcal{A}_0(\zeta) = 16\zeta^2\bar{\zeta}^2[\zeta^2 + (1 - \zeta)^2]$ contains precisely the transverse-photon splitting weight $[\zeta^2 + (1 - \zeta)^2]$ of the leading-order $\gamma_T^* \rightarrow q\bar{q}$ dijet.

3.2 The Wilson-line structures — exact algebra

By (P1), $y' \rightarrow y$ and $x' \rightarrow x$ as $\xi \rightarrow 0$. Insert (1.5)–(1.8) into $\mathbb{W}_{\bar{q}}$ and use unitarity, $S(a, a) = 1$, $Q(a, a, c, d) = S(c, d)$, $Q(a, b, c, c) = S(a, b)$:

$$\begin{aligned}
\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) &= Q(x, z, \bar{z}, \bar{x})Q(z, y, \bar{y}, \bar{z}) - S(x, z)S(z, y) - S(\bar{z}, \bar{x})S(\bar{y}, \bar{z}) + 1, \\
\mathcal{W}(x, y, y, \bar{x}, \bar{y}, \bar{z}) &= Q(x, y, \bar{z}, \bar{x})S(\bar{y}, \bar{z}) - S(x, y) - S(\bar{z}, \bar{x})S(\bar{y}, \bar{z}) + 1, \\
\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{y}) &= Q(x, z, \bar{y}, \bar{x})S(z, y) - S(x, z)S(z, y) - S(\bar{y}, \bar{x}) + 1, \\
\mathcal{W}(x, y, y, \bar{x}, \bar{y}, \bar{y}) &= Q(x, y, \bar{y}, \bar{x}) - S(x, y) - S(\bar{y}, \bar{x}) + 1.
\end{aligned}$$

Forming $\mathcal{W}_1 - \mathcal{W}_2 - \mathcal{W}_3 + \mathcal{W}_4$, **every** $S \cdot S$ term, every single S , and the four +1's cancel pairwise, leaving

$$\mathbb{W}_{\bar{q}}^{(0)} = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - Q(x, y, \bar{z}, \bar{x}) S(\bar{y}, \bar{z}) - Q(x, z, \bar{y}, \bar{x}) S(z, y) + Q(x, y, \bar{y}, \bar{x})$$

Identically for the quark channel, from (1.13)–(1.16) with $x' \rightarrow x$:

$$\mathbb{W}_q^{(0)} = Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z}) - Q(y, x, \bar{z}, \bar{y}) S(\bar{x}, \bar{z}) - Q(y, z, \bar{x}, \bar{y}) S(z, x) + Q(y, x, \bar{x}, \bar{y})$$

and for the interference, from (1.21)–(1.24) with $x' \rightarrow x, \bar{y}' \rightarrow \bar{y}$:

$$\mathbb{W}_{q\bar{q}}^{(0)} = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) - Q(x, y, \bar{y}, \bar{z}) S(\bar{z}, \bar{x}) - Q(x, z, \bar{y}, \bar{x}) S(z, y) + S(x, y) S(\bar{y}, \bar{x})$$

[Verified symbolically, checkw2.py; the same script confirms each combination vanishes at $\mathbf{z} \rightarrow \mathbf{y} / \mathbf{z} \rightarrow \mathbf{x}$, which is the statement of §2.2.]

The disappearance of the $-S_{q\bar{q}g} - S_{q\bar{q}g}^\dagger + 1$ pieces is worth noting: those are the "virtual" parts of $\mathcal{W}$, and they cancel between the four terms of the $\mathbb{W}$ combination. What remains is purely the **real** emission structure — which is exactly right, because the virtual half of the B-JIMWLK kernel must come from the one-loop $q\bar{q}$ (dijet) amplitude, not from the trijet. See §6.2.

3.3 The transverse kernels and the leading-order factor

Contract the free indices. With $\delta^{ij}\delta^{mn}$ surviving,

$$\frac{R^i Y^m \bar{R}^j \bar{Y}^n}{Y^2 \bar{Y}^2} \delta^{ij} \delta^{mn} = (R \cdot \bar{R}) \frac{Y \cdot \bar{Y}}{Y^2 \bar{Y}^2} \xrightarrow{\xi \rightarrow 0} (R_0 \cdot \bar{R}_0) \mathcal{K}_{q\bar{q}},$$

$$\frac{\tilde{R}^i X^m \tilde{\bar{R}}^j \tilde{\bar{X}}^n}{X^2 \tilde{X}^2} \delta^{ij} \delta^{mn} \xrightarrow{\xi \rightarrow 0} (R_0 \cdot \bar{R}_0) \mathcal{K}_{qq}, \quad \frac{\tilde{R}^i X^m \bar{R}^j \bar{Y}^n}{X^2 \bar{Y}^2} \delta^{ij} \delta^{mn} \xrightarrow{\xi \rightarrow 0} (R_0 \cdot \bar{R}_0) \mathcal{K}_{q\bar{q}},$$

by (P4)–(P5), and $\frac{K_1(Q\mathcal{D})K_1(Q\bar{\mathcal{D}})}{\mathcal{D}\bar{\mathcal{D}}} \rightarrow \frac{K_1(Q\mathcal{D}_0)K_1(Q\bar{\mathcal{D}}_0)}{\mathcal{D}_0\bar{\mathcal{D}}_0}$ by (P3).

3.4 Assembly

Putting §3.1–§3.3 together, and remembering the explicit – and the $2\Re$ in (1.17),

$$\mathcal{G}(0) = \mathcal{C} \mathcal{A}_0(\zeta) \int_{\Xi} (R_0 \cdot \bar{R}_0) \frac{K_1(Q\mathcal{D}_0)K_1(Q\bar{\mathcal{D}}_0)}{\mathcal{D}_0\bar{\mathcal{D}}_0} \left\{ \mathcal{K}_{q\bar{q}} \mathbb{W}_{\bar{q}}^{(0)} + \mathcal{K}_{qq} \mathbb{W}_q^{(0)} - 2\Re[\mathcal{K}_{q\bar{q}} \mathbb{W}_{q\bar{q}}^{(0)}] \right\}$$

$$\mathcal{A}_0(\zeta) = 16 \zeta^2 (1 - \zeta)^2 [\zeta^2 + (1 - \zeta)^2]$$

4. What the divergent coefficient is

4.1 It is the LO dijet times a colour kernel

The ξ -independent factor $\mathcal{A}_0(\zeta)(R_0 \cdot \bar{R}_0)K_1K_1/(\mathcal{D}_0\bar{\mathcal{D}}_0)$ is, up to the overall normalization, precisely the leading-order transverse-photon $\gamma_T^* + A \rightarrow q\bar{q}$ dijet integrand written at the four points $x, \bar{x}, y, \bar{y}$. Everything that depends on the gluon — $\mathbf{z}, \bar{\mathbf{z}}, k_3$ — sits in the curly bracket. That is the statement that the slow gluon has factorized off the hard process.

4.2 The kernel is the BK / B-JIMWLK emission kernel — the check

Write the eikonal emission amplitude off the $q\bar{q}$ dipole,

$$\mathcal{A}^m(\mathbf{z}) = \frac{(\mathbf{z} - \mathbf{x})^m}{(\mathbf{z} - \mathbf{x})^2} - \frac{(\mathbf{z} - \mathbf{y})^m}{(\mathbf{z} - \mathbf{y})^2}, \quad \bar{\mathcal{A}}^n(\bar{\mathbf{z}}) = \frac{(\bar{\mathbf{z}} - \bar{\mathbf{x}})^n}{(\bar{\mathbf{z}} - \bar{\mathbf{x}})^2} - \frac{(\bar{\mathbf{z}} - \bar{\mathbf{y}})^n}{(\bar{\mathbf{z}} - \bar{\mathbf{y}})^2}.$$

Then, **exactly, with no coincidence limit and no approximation**,

$$\mathcal{K}_{qq} + \mathcal{K}_{\bar{q}\bar{q}} - \mathcal{K}_{q\bar{q}} - \mathcal{K}_{\bar{q}q} = \mathcal{A} \cdot \bar{\mathcal{A}}, \quad \mathcal{K}_{\bar{q}q} \equiv \frac{(z - y) \cdot (\bar{z} - \bar{x})}{(z - y)^2 (\bar{z} - \bar{x})^2},$$

and $\mathcal{K}_{\bar{q}q}$ is exactly what the complex conjugate half of the $-2\Re[\mathcal{K}_{q\bar{q}} \mathbb{W}_{q\bar{q}}^{(0)}]$ term supplies. So the curly bracket of §3.4 is the modulus-squared of a single eikonal emission amplitude, weighted by colour. At coincident transverse points $\bar{x} = x$, $\bar{y} = y$, $\bar{z} = z$ this becomes

$$|\mathcal{A}|^2 = \frac{1}{(z - x)^2} + \frac{1}{(z - y)^2} - \frac{2(z - x) \cdot (z - y)}{(z - x)^2 (z - y)^2} = \frac{(x - y)^2}{(z - x)^2 (z - y)^2},$$

the Balitsky–Kovchegov dipole kernel. [Both identities verified symbolically; see §8.2.]

Why this is a strong check. Three independently computed contributions, with three different spinor structures (1.2), (1.10), (1.18) and three different Wilson-line combinations (1.5)–(1.8), (1.13)–(1.16), (1.21)–(1.24), conspire to produce a *single* common weight $\mathcal{A}_0(\zeta)$ (§3.1) multiplying a kernel combination that is a perfect square. If any of the limits in §3.1 or §3.2 were wrong, or if (P3)–(P5) failed, this structure would not appear.

What this does *not* yet say. $\mathcal{G}(0)$ is the **real-emission** part of the kernel only. Getting the full BK right-hand side $[S(x, z)S(z, y) - S(x, y)]$ requires two further ingredients that are not in your trijet: (a) large- N_c factorization of the quadrupoles in $\mathbb{W}^{(0)}$, and (b) the virtual term $-S(x, y)$, which comes from the one-loop $q\bar{q}$ amplitude, *not* from a three-parton final state. See §6.2. For the trijet observable itself this is not a limitation — the real part is the whole answer — but it matters when you write the evolution equation.

5. The separation

5.1 Pointwise (unintegrated) form

This is the "usual way": add and subtract the pole. No cutoff, no region decomposition, no auxiliary parameter.

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} = \underbrace{\frac{\mathcal{G}(0)}{\xi}}_{\left[\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} \right]_{\text{div}}} + \underbrace{\frac{\mathcal{G}(\xi) - \mathcal{G}(0)}{\xi}}_{\left[\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} \right]_{\text{fin}}}$$

The second term is finite at $\xi = 0$ because $\mathcal{G}(\xi) - \mathcal{G}(0) = \mathcal{O}(\xi)$; §5.3 gives the coefficient explicitly. The first term carries the entire singularity.

5.2 Distributional (integrated) form

If you afterwards integrate over the gluon's longitudinal fraction with the regulator $\xi > \xi_0 = \Lambda^+/q^+$ and a rapidity factorization scale ξ_f , the same statement reads

$$\int_{\xi_0}^{\xi_f} \frac{d\xi}{\xi} \mathcal{G}(\xi) = \underbrace{\mathcal{G}(0) \ln \frac{\xi_f}{\xi_0}}_{\text{divergent}} + \underbrace{\int_0^{\xi_f} \frac{d\xi}{\xi} [\mathcal{G}(\xi) - \mathcal{G}(0)]}_{\text{finite, } \xi_0\text{-independent}},$$

i.e. the plus-prescription of §0. The $\ln(1/\xi_0)$ is removed by defining the Wilson-line correlators S, Q at the rapidity $Y_f = \ln(1/\xi_f)$; see §6.

5.3 The finite part, written out

Since only the $\delta^{ij}\delta^{mn}$ Regular $\times$ Regular structure survives at $\xi = 0$, the subtraction touches only that term. Explicitly, the finite part is the sum of five pieces:

(i) Subtracted Regular $\times$ Regular, $\delta\delta$ — antiquark channel. Using the exact $\mathcal{A}_{\bar{q}}$ of §3.1,

$$\frac{\mathcal{A}_{\bar{q}}(\zeta, \xi) - \mathcal{A}_0(\zeta)}{\xi} = 8\zeta^2[\zeta^2 + \bar{\zeta}^2](\xi - 2\bar{\zeta}) \xrightarrow{\xi \rightarrow 0} -16\zeta^2\bar{\zeta}[\zeta^2 + \bar{\zeta}^2],$$

multiplying $\int_{\Xi} (R \cdot \bar{R}) \frac{Y \cdot \bar{Y}}{Y^2 \bar{Y}^2} \frac{K_1(Q\mathcal{D})K_1(Q\bar{\mathcal{D}})}{\mathcal{D}\bar{\mathcal{D}}} \mathbb{W}_{\bar{q}}$, plus the piece in which $\mathcal{A}_0$ multiplies the difference of the transverse structures,

$$\frac{\mathcal{A}_0(\zeta)}{\xi} \left[(R \cdot \bar{R}) \frac{Y \cdot \bar{Y}}{Y^2 \bar{Y}^2} \frac{K_1 K_1}{\mathcal{D}\bar{\mathcal{D}}} \mathbb{W}_{\bar{q}} - (R_0 \cdot \bar{R}_0) \mathcal{K}_{\bar{q}\bar{q}} \frac{K_1(Q\mathcal{D}_0)K_1(Q\bar{\mathcal{D}}_0)}{\mathcal{D}_0 \bar{\mathcal{D}}_0} \mathbb{W}_{\bar{q}}^{(0)} \right],$$

which is finite because each factor differs from its $\xi = 0$ value by $\mathcal{O}(\xi)$ — for $\mathbb{W}_{\bar{q}}$ this follows from (P2), $y' - y = \frac{\xi}{\eta + \xi}(z - y)$.

(ii), (iii) The same for the quark and interference channels, with

$$\begin{aligned} \frac{\mathcal{A}_q(\zeta, \xi) - \mathcal{A}_0(\zeta)}{\xi} &\xrightarrow{\xi \rightarrow 0} -16\zeta\bar{\zeta}(10\zeta^3 - 14\zeta^2 + 7\zeta - 1), \\ \frac{\mathcal{A}_{q\bar{q}}(\zeta, \xi) - \mathcal{A}_0(\zeta)}{\xi} &\xrightarrow{\xi \rightarrow 0} -8\zeta\bar{\zeta}(12\zeta^3 - 16\zeta^2 + 8\zeta - 1). \end{aligned}$$

(iv) The $\epsilon^{ij}\epsilon^{mn}$ terms, unsubtracted. They already carry an explicit ξ , so $\frac{1}{\xi} \times \mathcal{O}(\xi)$ is finite with no subtraction at all:

$$\begin{aligned} &-\frac{8\zeta^2(2\zeta - 1)(2\bar{\zeta} - \xi)}{1} (R \wedge \bar{R}) \frac{Y \wedge \bar{Y}}{Y^2 \bar{Y}^2} \text{ [antiquark]}, \quad -8\eta^2(2\eta - 1)(2(1 - \eta) - \xi) (\tilde{R} \wedge \bar{\tilde{R}}) \frac{X \wedge \bar{X}}{X^2 \bar{X}^2} \text{ [quark]}, \\ &+8\eta\zeta [1 - 2\zeta - \xi]^2 (\tilde{R} \wedge \bar{R}) \frac{X \wedge \bar{Y}}{X^2 \bar{Y}^2} \text{ [interference]}, \end{aligned}$$

with $a \wedge b \equiv \epsilon^{ij}a^i b^j$.

(v) Regular $\times$ Instantaneous and Instantaneous $\times$ Instantaneous, unsubtracted. Likewise finite after the $1/\xi$, since (1.3),(1.11),(1.19) are $\mathcal{O}(\xi)$ and (1.4),(1.12) are $\mathcal{O}(\xi^2)$, while (1.20) vanishes identically. For example the antiquark Reg $\times$ Inst term contributes

$$\frac{8\zeta^2\eta}{1 - \zeta} \left[\zeta(2(1 - \zeta) - \xi) \frac{(R \cdot \bar{Y})(\bar{R} \cdot Y)}{Y^2 \bar{Y}^2} - \xi(1 - \zeta) \frac{(R \wedge \bar{Y})(\bar{R} \wedge Y)}{Y^2 \bar{Y}^2} \right] \left[\frac{Y^2}{R \cdot Y} + \frac{\bar{Y}^2}{\bar{R} \cdot \bar{Y}} \right] \frac{K_1 K_1}{\mathcal{D}\bar{\mathcal{D}}} \mathbb{W}_{\bar{q}},$$

with no $1/\xi$ left over.

Practical form. Numerically the cleanest implementation of (i)–(iii) is not to expand at all, but to evaluate $\frac{1}{\xi}[\mathcal{G}(\xi) - \mathcal{G}(0)]$ directly: build $\mathcal{G}(\xi)$ from your expressions as they stand, build $\mathcal{G}(0)$ from the boxed formula of §3.4, and take the difference before dividing by ξ . The cancellation is exact and no expansion in ξ is needed.

6. What happens to the divergent part

6.1 Absorption into the target

The divergent piece $\mathcal{G}(0) \ln(\xi_f/\xi_0)$ is removed by declaring that the correlators S and Q appearing in your $\mathcal{W}$'s are not fixed objects but are defined at a rapidity $Y_f = \ln(1/\xi_f)$, and evolve:

$$S_{Y_f} = S_{Y_0} + \ln\left(\frac{\xi_f}{\xi_0}\right) [\mathcal{H} \otimes S]_{Y_0} + \mathcal{O}(\alpha_s^2), \quad \frac{\partial}{\partial Y_f} \langle \cdot \rangle_{Y_f} = \mathcal{H}_{\text{JIMWLK}} \otimes \langle \cdot \rangle_{Y_f},$$

with $Y_0 = \ln(1/\xi_0)$ and, at large N_c for the dipole,

$$\frac{\partial S(x, y)}{\partial Y} = \frac{\bar{\alpha}_s}{2\pi} \int d^2 z \frac{(\mathbf{x} - \mathbf{y})^2}{(\mathbf{z} - \mathbf{x})^2 (\mathbf{z} - \mathbf{y})^2} [S(x, z) S(z, y) - S(x, y)], \quad \bar{\alpha}_s = \frac{\alpha_s N_c}{\pi}.$$

The requirement that the physical cross-section be independent of ξ_f at this order is precisely this equation — and its right-hand side is what §4.2 showed $\mathcal{G}(0)$ to be.

6.2 The virtual terms: why $\mathbb{W}^{(0)}$ contains only real emission

In §3.2 the $-S_{q\bar{q}g} - S_{q\bar{q}g}^\dagger + 1$ parts of $\mathcal{W}$ cancelled completely, leaving only quadrupole and dipole *products*. This is as it must be: the $-S(x, y)$ (virtual) term of the BK equation cannot come from a three-parton final state, since the gluon is measured. It comes from the one-loop correction to the *dijet* ($\gamma^* \rightarrow q\bar{q}$) amplitude.

So: **the trijet alone does not give you BK or JIMWLK**. It gives you the real half of the kernel. When you use the divergent part of your trijet to define the evolution, you must add the $\xi \rightarrow 0$ limit of the one-loop dijet to supply the virtual half; only the sum is probability-conserving (equivalently, only the sum has a $\mathbf{z}$ -integral that converges at large $|\mathbf{z}|$ after the correlators are replaced by their dilute-limit expansion). For the trijet observable itself this is not a problem — the real part is the whole answer — but it matters the moment you write down an evolution equation.

6.3 Lifetime ordering — a refinement of $\mathcal{G}(0)$, not of the method

Setting $\xi = 0$ inside $\mathcal{D}(x, y, z, \zeta, \xi)$ was step (P3). If, as in the standard LCPT form,

$$\mathcal{D}^2 = \zeta \eta (\mathbf{x} - \mathbf{y})^2 + \zeta \xi (\mathbf{x} - \mathbf{z})^2 + \eta \xi (\mathbf{y} - \mathbf{z})^2,$$

then dropping the ξ -terms is legitimate only where $\xi (\mathbf{y} - \mathbf{z})^2 \lesssim \zeta \eta (\mathbf{x} - \mathbf{y})^2$ — because $\mathbf{z}$ is integrated over, and however small ξ is there is always a region of large $|\mathbf{y} - \mathbf{z}|$ where the neglected term dominates. In momentum variables that condition is

$$\frac{\xi}{k_{3\perp}^2} \lesssim \frac{\zeta \eta}{Q^2} \quad \Longleftrightarrow \quad \frac{1}{k_3^+} \lesssim \frac{1}{k_f^+},$$

i.e. the emitted gluon must be **shorter-lived than the parent $q\bar{q}$ fluctuation**. The consequence for your separation is *not* a change of method — the subtraction of §5 is exact as it stands — but a refinement of what you feed into the evolution: the kernel should carry the kinematic constraint

$$\Theta\left(Y_f - Y - \ln\left(\frac{\min[(\mathbf{z} - \mathbf{x})^2, (\mathbf{z} - \mathbf{y})^2]}{(\mathbf{x} - \mathbf{y})^2}\right)\right),$$

and the phase space thereby removed from the evolution must be returned to the impact factor. That returned piece is a Sudakov double logarithm with a negative coefficient; it is $\mathcal{O}(\alpha_s)$ and does not affect the divergence separation at the order you are working. It matters if and when you integrate out the gluon and look at back-to-back kinematics.

7. If you subsequently integrate the gluon out

Your original aim was "finite real trijet plus finite virtual dijet". The subtraction above delivers the first half of that, exactly:

- **The subtracted trijet** $\frac{1}{\xi}[\mathcal{G}(\xi) - \mathcal{G}(0)]$ is integrable in ξ down to $\xi = 0$. It can be integrated over any part of the gluon phase space without a rapidity regulator.
- **The collinear region.** Once you integrate over the relative momentum of, say, the antiquark and the gluon, the boundary of §2.3 turns into a genuine logarithm. In the fully differential trijet it is absent; in a dijet observable it must be handled by the jet definition, and it cancels against the collinear pole of the one-loop dijet by Kinoshita–Lee–Nauenberg. That cancellation is between the *integrated* real trijet and the virtual dijet, and is a separate step from the rapidity subtraction done here.
- **The virtual dijet** is an independent input: it supplies (a) the virtual half of the JIMWLK kernel (§6.2), and (b) the collinear pole that cancels the one just described.

8. Summary of the analytic result

Divergent part — the entire singularity of the LO trijet:

$$\left[\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} \right]_{\text{div}} = \frac{1}{\xi} \mathcal{C} 16 \zeta^2 (1 - \zeta)^2 [\zeta^2 + (1 - \zeta)^2] \\ \times \int_{x, y, z, \bar{x}, \bar{y}, \bar{z}} e^{-ik_1 \cdot (x - \bar{x})} e^{-ik_2 \cdot (y - \bar{y})} e^{-ik_3 \cdot (z - \bar{z})} (R_0 \cdot \bar{R}_0) \frac{K_1(Q\mathcal{D}_0) K_1(Q\bar{\mathcal{D}}_0)}{\mathcal{D}_0 \bar{\mathcal{D}}_0} \\ \times \left\{ \frac{(z - y) \cdot (\bar{z} - \bar{y})}{(z - y)^2 (\bar{z} - \bar{y})^2} \mathbb{W}_q^{(0)} + \frac{(z - x) \cdot (\bar{z} - \bar{x})}{(z - x)^2 (\bar{z} - \bar{x})^2} \mathbb{W}_q^{(0)} - 2\Re \left[\frac{(z - x) \cdot (\bar{z} - \bar{y})}{(z - x)^2 (\bar{z} - \bar{y})^2} \mathbb{W}_{q\bar{q}}^{(0)} \right] \right\}$$

with $\mathbb{W}_q^{(0)}$, $\mathbb{W}_q^{(0)}$, $\mathbb{W}_{q\bar{q}}^{(0)}$ the three quadrupole structures of §3.2. Integrated, this is $\mathcal{G}(0) \ln(\xi_f/\xi_0)$, absorbed by B-JIMWLK (large N_c : BK) evolution of S, Q to $Y_f = \ln(1/\xi_f)$.

Finite part — your cross-section minus the above, term by term:

$$\left[\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} \right]_{\text{fin}} = \frac{\mathcal{C}}{\xi} \left\{ [\mathcal{I}_{\bar{q}} + \mathcal{I}_q - 2\Re \mathcal{I}_{q\bar{q}}](\xi) - [\mathcal{I}_{\bar{q}} + \mathcal{I}_q - 2\Re \mathcal{I}_{q\bar{q}}](0) \right\}$$

with $\mathcal{I}$ as defined in §1.2. It is regular at $\xi = 0$, with the limits given in §5.3; it requires no regulator; and it contains, unsubtracted, all the $\epsilon^{ij} \epsilon^{mn}$, Regular×Instantaneous and Instantaneous×Instantaneous structures, each of which is already $\mathcal{O}(\xi)$ or smaller.

8.1 What was assumed, and how to check it

#	Assumption	Used in	How to check
(P1)	$y' \rightarrow y, x' \rightarrow x$ as $\xi \rightarrow 0$	§3.2	Immediate from $\mathbf{y}' = \frac{\eta \mathbf{y} + \xi \mathbf{z}}{\eta + \xi}$.
(P2)	$y' \rightarrow y$ as $z \rightarrow y$, linearly	§2.2 (UV finiteness)	Same formula.
(P3)	$\mathcal{D} _{\xi=0}$ is the same in the quark and antiquark channels	§3.3	Required for the common prefactor $\mathcal{A}_0$; holds for the standard $\mathcal{D}^2 = \zeta \eta r_{xy}^2 + \zeta \xi r_{xz}^2 + \eta \xi r_{yz}^2$.
(P4)	$R _{\xi=0} = \bar{R} _{\xi=0}$	§3.3	Required for the three kernels to combine; verified <i>a posteriori</i> by §4.2.
(P5)	$Y^m/Y^2 \rightarrow (z - y)^m/(z - y)^2$, $X^m/X^2 \rightarrow (z - x)^m/(z - x)^2$	§3.3	Required for the BK kernel; if $\mathbf{Y} = \mathbf{z} - \mathbf{y}'$ instead of $\mathbf{z} - \mathbf{y}$, the two differ by $\eta/(\eta + \xi) \rightarrow 1$ and the limit is unchanged.

Nothing in §3.1 (spinor structures) or §3.2 (Wilson lines) uses (P3)–(P5); those two sections are unconditional consequences of the expressions in `Trijet_LO.pdf`.

8.2 Verification log

Statement	Method	Result
$\mathcal{A}_{\bar{q}}(\zeta, 0) = \mathcal{A}_q(\zeta, 0) = \mathcal{A}_{q\bar{q}}(\zeta, 0) = 16\zeta^2\bar{\zeta}^2[\zeta^2 + \bar{\zeta}^2]$	sympy, checkA.py	all three equal — exact
All $\epsilon\epsilon$, Reg $\times$ Inst, Inst $\times$ Inst structures $\rightarrow 0$ as $\xi \rightarrow 0$	sympy, checkW.py	confirmed; $\mathcal{O}(\xi)$, $\mathcal{O}(\xi)$, $\mathcal{O}(\xi^2)$
$\mathcal{A}(\zeta, \xi) - \mathcal{A}(\zeta, 0) = \mathcal{O}(\xi)$ in all three channels	sympy series	confirmed — the subtraction is finite
The S 's and $+1$'s cancel in $\mathbb{W}_{\bar{q}}^{(0)}, \mathbb{W}_q^{(0)}, \mathbb{W}_{q\bar{q}}^{(0)}$	symbolic, checkW2.py	confirmed; results as in §3.2
$\mathbb{W}_{\bar{q}} \rightarrow 0$ at $z \rightarrow y$ and at $\bar{z} \rightarrow \bar{y}$; $\mathbb{W}_{q\bar{q}} \rightarrow 0$ at $z \rightarrow x$	symbolic	identically zero — UV finiteness
$\mathcal{K}_{\bar{q}\bar{q}} + \mathcal{K}_{qq} - 2\mathcal{K}_{q\bar{q}} = (x - y)^2 / [(z - x)^2(z - y)^2]$	symbolic	exact — the BK kernel